Supplementary Material for

Friendship shapes children's predictions of emotion responses following conflicting outcomes

Methods

Participants

Sample Size

The preregistered target sample was 144 children, with recruitment continuing until at least 48 children were enrolled in each of three age groups (4-5, 6-8, and 9-10 years). The final sample included 151 children (48, 55, and 48 children in each age group, respectively). Thus, although the final sample exceeded the anticipated total, recruitment followed the preregistered stopping rule.

Demographics

Participants were 151 U.S. children ages 4 to 10 (M_{age} = 7.62 years; SD_{age} = 2.10 years; 81 girls). Seventy-two children were identified by their caregiver as White, 25 as Asian, 15 as Black, 1 as American Indian/Alaskan Native, 32 were identified as belonging to two or more races, and 6 caregivers declined to answer. 11 children were identified by their caregiver as Hispanic/Latinx.

One hundred twenty-one children came from families where at least one caregiver reported a college degree or higher, 9 with an associate's degree, 11 with some college completion, 2 with a high school diploma or equivalent, 1 without a high school diploma, and 7 declined to answer. The median household income was between $100,000 and $149,000.

Exclusion Information

Exclusion criteria were preregistered prior to data collection. Individual stories were excluded if participants failed an attention check after the story was repeated, provided an explanation that contradicted the story content, or experienced an experimenter error. Participants were excluded from the full sample and replaced if two or more of their stories were excluded, regardless of the reason. In practice, no excluded participants met multiple exclusion criteria (**Table S1**).

Outcome Type	**Relationship Type**	**Exclusion Reason**	***n***
Negative	Friends	Contradicted Story Content	7
		Failed Attention Check	1
		Experimenter Error	0
	Rivals	Contradicted Story Content	3
		Failed Attention Check	0
		Experimenter Error	1
Positive	Friends	Contradicted Story Content	12
		Failed Attention Check	2
		Experimenter Error	2
	Rivals	Contradicted Story Content	10
		Failed Attention Check	2
		Experimenter Error	1

Table S1. Story-level exclusions by relationship type, outcome type, and exclusion reason.

Measures

Empathy and Theory of Mind Scale (EToMS)

Caregivers completed the Empathy and Theory of Mind Scale (EToMS; Wang & Wang, 2015), which assesses three dimensions of children's social cognition: Empathy, Nice Theory of Mind (ToM), and Nasty ToM. The Empathy subscale assesses children's tendency to respond

empathically to others (5 items), whereas the Nice ToM (5 items) and Nasty ToM (6 items) subscales assess children's use of mental-state understanding in prosocial and antisocial contexts, respectively. Caregivers were given a list of behaviors and rated the frequency in which their child engaged in each behavior on a scale of never (1) to always (5). Consistent with our preregistration, mean scores were calculated separately for each subscale, with higher scores indicating greater caregiver-reported abilities or tendencies in the corresponding domain. Subscale scores were mean-centered before inclusion in the exploratory analyses. Each subscale was examined in a separate model to assess its association with children's emotion predictions.

Results

Additional Emotion Prediction Analyses

Preregistered Pairwise Comparisons

In addition to the relationship comparisons reported in the main text, we conducted preregistered comparisons of emotion scores following positive versus negative outcomes within each relationship type. Bonferroni-corrected comparisons revealed that children predicted more positive emotions following positive than negative outcomes both when the characters were friends, $t(428) = 18.87$, $p < .001$, 95% CI [2.79, 3.44], $d = 2.27$, and when they were rivals, $t(420) = 28.91$, $p < .001$, [4.39, 5.03], $d = 3.43$. Thus, although the magnitude of children's emotion predictions varied by relationship type, children generally expected the protagonist's emotions to correspond to their own outcome.

Separate Happy and Sad Ratings

To further characterize children's emotion predictions, we examined their happiness and sadness ratings separately rather than combining them into a composite score. A model including relationship, outcome, emotion (happiness vs. sadness), age, and their interactions revealed a

significant relationship x outcome x emotion interaction, $\chi^2(1) = 72.44$, $p < .001$. Follow-up comparisons showed that children predicted less sadness ($b = 0.53$, $t(972) = 5.81$, $p < .001$) and more happiness ($b = 0.30$, $t(972) = 3.25$, $p = .001$) for friends than for rivals after negative outcomes. Conversely, following positive outcomes, children predicted less happiness ($b = -0.37$, $t(991) = -3.98$, $p < .001$) and more sadness ($b = -0.39$, $t(991) = -4.15$, $p < .001$) for friends than rivals. Although the four-way interaction with age was significant, $\chi^2(1) = 7.76$, $p = .005$, we did not pursue age-specific comparisons because we lacked sufficient power to characterize this higher-order interaction reliably. These findings indicate that the relationship-dependent pattern observed in composite scores was also reflected in children's separate emotion ratings.

Story Context

To test whether children's emotion predictions differed across story contexts, we ran an additional mixed-effects analysis including relationship, outcome, context (competition vs. limited resources), and their interactions, with participants as a random intercept. There was no significant main effect of context, $\chi^2(1) = 2.29$, $p = .131$, nor did context interact significantly with relationship type, $\chi^2(1) = 0.09$, $p = .763$, outcome type, $\chi^2(1) = 0.04$, $p = .843$, or their interaction, $\chi^2(1) = 0.41$, $p = .523$. The relationship x outcome interaction remained significant, $\chi^2(1) = 44.92$, $p < .001$, suggesting that the primary pattern was similar across competition and limited-resource scenarios.

Presentation Order

To examine whether presentation order influenced children's emotion predictions, we conducted an additional mixed-effects analysis including relationship, outcome, their interaction, and the relationship, outcome, and context presented in the first story, with participant included as a random intercept. There was a significant effect of outcome presentation order, $\chi^2(1) =$

11.04, $p < .001$. Children who first encountered a negative outcome provided somewhat more positive composite emotion ratings overall ($M = 0.25$, $SD = 2.36$) than children who first encountered a positive outcome ($M = -0.03$, $SD = 2.50$). Neither relationship presentation order, $\chi^2(1) = 1.52$, $p = .218$, nor context presentation order, $\chi^2(1) = 0.53$, $p = .469$, significantly predicted emotion ratings. Importantly, the relationship × outcome interaction remained significant, $\chi^2(1) = 45.70$, $p < .001$, indicating that the primary finding persisted after accounting for presentation order.

Explanations

Explanations Coding

Following each emotion rating, children were asked to explain their predictions. Responses were transcribed and assigned to one of eight mutually exclusive categories based on the information children used to justify their judgments (**Table S2**). The categories distinguished explanations referencing the protagonist's own outcome, the other character's outcome, the relationship between the characters, and other considerations relevant to the predicted emotion.

Category	**Definition**	**Examples**
Own Outcome	References what happened to the target.	"He won the race." "She lost." "He didn't get the cupcake."
Other Outcome	References what happened to the other character.	"His friend lost." "Her friend won." "Morgan didn't get the cupcake."
Relationship	References the friendship, dislike, or other relationship between the characters as relevant to the emotion.	"Morgan is his friend." "Quinn isn't her friend." "They're not friends."
Other-Directed Concern	Explains the target's emotion in terms of caring about,	"Happy for his friend." "She feels bad that her friend

	feeling for, or emotionally responding to the other character's outcome or welfare.	didn't win." "He doesn't care that the other boy lost." "She wanted her friend to win too."
Integrated Reasoning	Integrates both outcome and relationship to explain the target's emotional response.	"He lost, but his friend won." "She won, but they're best friends, so she doesn't feel very happy." "Morgan is not his friend, so he feels very happy that he got the cupcake."
Desire or Value	Explains emotion in terms of how much the target wants, likes, or values the outcome/resource.	"He really wanted to win." "She really wanted that sticker."
Mitigation or Reappraisal	Reduces, reframes, compensates for, or proposes a way to repair the emotional significance of an outcome.	"It's just a game." "They could do a rematch." "There will be other stickers." "He can learn from his mistakes."
Other	Does not provide an interpretable explanation for the predicted emotion.	"I don't know."

Table S2. Coding categories for children's explanations.

Across 1,107 coded explanations, children most frequently referenced the protagonist's own outcome (69.2%). Integrated Reasoning accounted for 8.6% of responses, followed by Desire or Value (5.9%), Other (5.7%), Other Outcome (4.6%), Mitigation or Reappraisal (2.9%), Other-Directed Concern (1.9%), and Relationship (1.3%).

We additionally distinguished between explanations for emotions that were congruent versus incongruent with the protagonist's own outcome. Congruent explanations involved happiness after a positive outcome or sadness after a negative outcome. Incongruent explanations concerned sadness following a positive outcome or happiness following a negative outcome.

Figure S1 presents the distribution of explanation categories across relationship and emotion-congruence conditions.

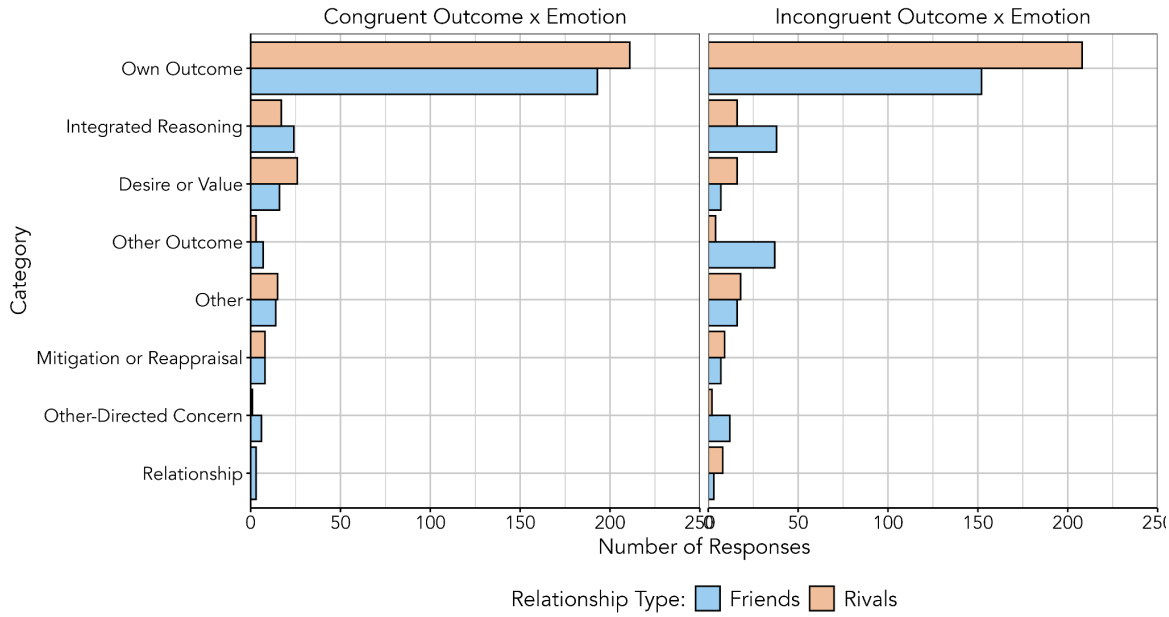


Figure S1. Distribution of children's explanations by relationship type and emotion congruence. Number of explanations assigned to each of the eight coding categories, separately for friends (blue) and rivals (orange), when the predicted emotion was congruent (left panel) or incongruent (right panel) with the protagonist's own outcome. Congruent emotions included happiness following a positive outcome and sadness following a negative outcome; incongruent emotions included sadness following a positive outcome and happiness following a negative outcome.

Socially Relevant Explanations

We were particularly interested in whether children referenced social information when explaining emotions that were incongruent with the protagonist's own outcome. Although happiness following success and sadness following failure can be explained by the protagonist's own outcome, explaining sadness following success or happiness following failure may require considering additional information, including the other character's outcome or the relationship between the characters. For the exploratory analysis reported in the main manuscript, explanations were classified according to whether they referenced socially relevant information.

Responses coded as Relationship, Other-Directed Concern, or Integrated Reasoning were classified as socially relevant (1); all remaining categories were classified as not socially relevant (0). The mixed-effects logistic regression reported in the main manuscript predicted whether explanations referenced socially relevant information from relationship type, emotion congruence, and their interaction, with participant included as a random intercept.

Exploratory Individual Differences

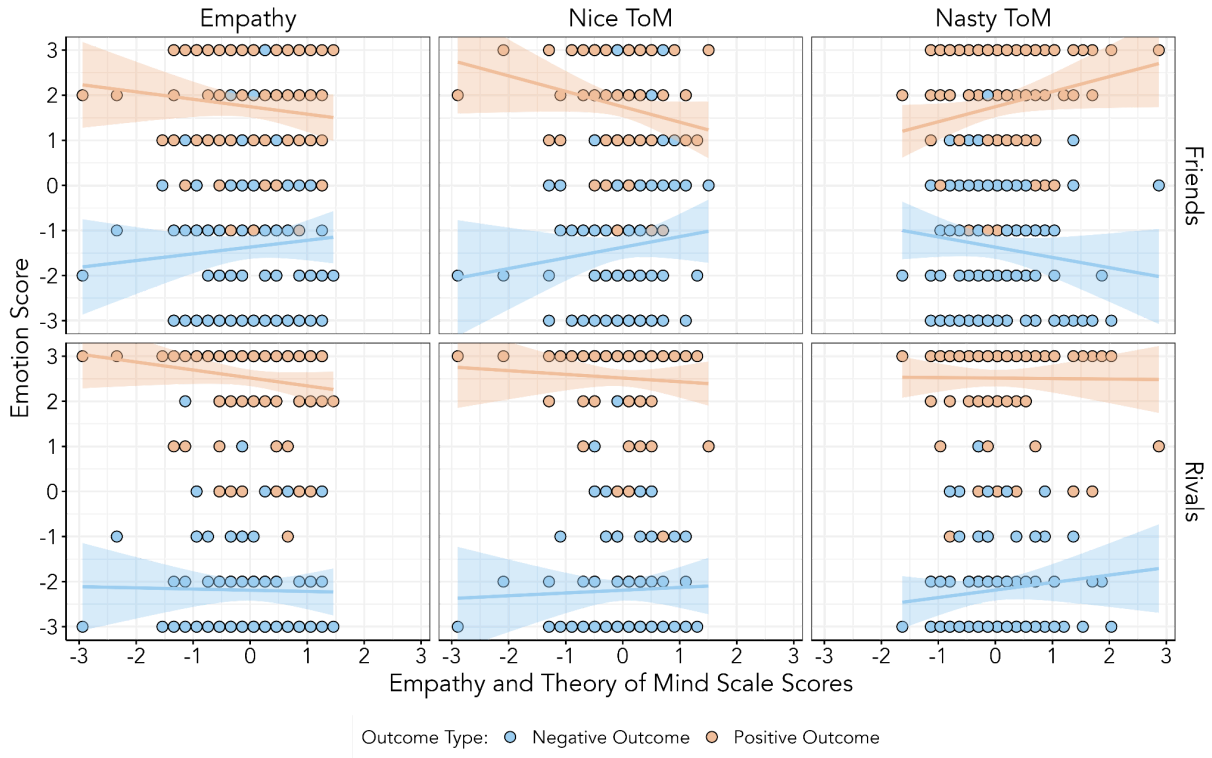


Figure S2. Associations between caregiver-reported empathy and theory of mind and children's emotion predictions. Children's composite emotion ratings as a function of caregiver-reported Empathy (left), Nice Theory of Mind (middle), and Nasty Theory of Mind (right), separately for friends (top) and rivals (bottom). Blue and orange indicate negative and positive outcomes, respectively. Points represent individual observations, solid lines show fitted regression estimates, and shaded bands indicate 95% confidence intervals. Analyses were exploratory.